Local Pachamama Beliefs and the Expansion of Cooperation among the Amazonian Kichwa

Connor Wood[a,b1]
Lexi Carver[c]
Daniel Major-Smith[a,d]
Benjamin Grant Purzycki[a]

[a]Aarhus University
[b] Center for Mind and Culture
[c] Independent researcher
[d] University of Bristol

Abstract: Religious beliefs regarding deities' punishment and knowledge breadth may facilitate the expansion of cooperative behaviour towards distant co-religionists. However, most previous research in this area has targeted so-called "moralistic" religious traditions (e.g., Christianity, Buddhism), and thus the extent to which these findings apply to other religious traditions is unclear. Here, we aimed to explore this question among the Kichwa, an indigenous group in Amazonian Ecuador, focusing on religious traditions regarding a pan-Quechua nature spirit, *Pachamama*. Our estimand of interest was *the joint causal effect of beliefs about Pachamama's punishment and omniscience on cooperation with distant Pachamama co-religionists*, with cooperation assessed via two experimental games (the Random Allocation Game and Dictator Game). Across all models and sensitivity analyses to different causal assumptions regarding the structure of confounding, we found little evidence for a relationship — either positive or negative — between Pachamama beliefs and cooperation. Assuming a causal interpretation, these results suggest that, in this population, beliefs regarding local deities do not motivate cooperative behaviour towards distant co-religionists.

[1] Contact authors' emails: Connor Wood (connor.wood@cas.au.dk); Daniel Major-Smith (dan.major-smith@cas.au.dk; dan.smith@bristol.ac.uk); Benjamin Grant Purzycki (bgpurzycki@cas.au.dk)

1. Background

1.1. Theory

For generations, thinkers have probed the relationship between religion and cooperation, variously construed (Durkheim, 2001; Malinowski, 1992; Rappaport, 1999). Recently, a newer body of research attends to the role that gods' knowledge, punishment, and moral concern may play in cooperative relationships. Specifically, some posit that there should be a positive relationship between cooperation and the degree to which gods are believed to know and care about, and punish, antisocial behavior (Johnson, 2015; Norenzayan, 2013a). One variant of this theory posits that humans evolved a sensitivity toward supernatural punishment because it reduces the costs of secular punishment for engaging in self-interested behavior (Johnson, 2005). Another variant posits that, because of the prosocial effects of such beliefs, traditions that espouse them will attract followers, thus facilitating broader cooperation, in turn allowing some groups to outcompete others in a cultural evolutionary scenario (Norenzayan et al., 2016). This cycle may thus partially account for the global dominance of the so-called "prosocial religions" such as Christianity and Islam. One implication of this theory is that, while such prosocial religions contribute to the expansion of cooperation, the cooperation promoted by smaller, "indigenous" or "local" traditions is presumably restricted to local communities. In other words, "world" religions such as Christianity and Islam may scale up cooperation to very wide scales and thus make it largely anonymous, whereas "local" traditions are exactly that: constrained to facilitate local, small-scale, and primarily face-to-face interactions.

The empirical study of the relationship between religion and cooperation is generally mixed (Billingsley et al., 2018; Gomes & McCullough, 2015; Shariff et al., 2016; Van Elk et al., 2015), but the specific work on culturally postulated beliefs and cooperation is more robust. While cross-cultural work shows that beliefs in supernatural punishment and the breadth of gods' knowledge are positively associated with fairness and generosity beyond parochial boundaries (Lang et al., 2019; Purzycki et al., 2016), the degree to which gods are thought to care about morality is not (Bendixen et al., 2023). In fact, across individual- and group-level studies, the association between gods and moral concern has been shown to be cross-culturally more common than previously claimed, extending even to small-scale societies in which cooperation is restricted to immediate groups (Lightner et al., 2023; Purzycki, 2011, 2025b; Purzycki, Willard, et al., 2022; Singh et al., 2021; Townsend et al., 2020). Furthermore, previous studies could not compare so-called "moralistic traditions" whose gods who give "specific support to human morality" such as fairness and harm avoidance (Swanson, 1960, p. 210) with traditions whose gods are more associated with other affairs, such as over-exploitation of resources, ritual, etiquette, and so forth (Purzycki & McNamara, 2016). Here, we examine the relationship between such "local" beliefs and cooperation among a community of Ecuadorian Kichwa.

1.2. Ethnographic Background: Lowland Kichwa

With a population of around 125,000, the lowland Kichwa (Quichua) are the largest of 11 indigenous minority ethnicities in Amazonian Ecuador (Heredia-R et al., 2020; Vasco et al., 2018). The ancestral Kichwa language is a member of the Quechua dialect continuum, with limited mutual intelligibility between lowland and high sierra Kichwa dialects within Ecuador (Grzech et al., 2019). Because of this, the lowland Kichwa have been stereotyped as recent Andean migrants into the Amazon, but comparative evidence instead indicates that the lowland Kichwa are the ethnic descendants of integration between precontact groups such as the Shuar, Achuar, and Quijo in the wake of the Spanish conquest and the dissolution of the Incan Empire (Heredia-R et al., 2020; Uzendoski & Whitten, 2014). Nevertheless, assumptions about an Andean origin have frequently been used to deny Kichwa land rights in the Amazon (Uzendoski & Whitten, 2014). Today, Spanish use is rapidly increasing, with only a minority of elders remaining Kichwa monoglots and most younger and middle-aged people either monolingual in or preferring Spanish (Grzech et al., 2019).

The lowland Kichwa are strongly tied, economically and ideologically, to the Amazon rainforest. Kichwa tend to devote a smaller portion of their land to agriculture or pasturage than mestizo colonists or other nearby Amazonian ethnic groups such as the Shuar, despite having larger households and larger average landholdings (Vasco et al., 2018). Households thus often have an abundance of land dominated by unmodified rainforest (Davis et al., 2017). Market integration has increased since the 1970s, with more extensive road networks connecting Kichwa territories to the rest of Ecuador. Accordingly, cash crops such as cacao and coffee have grown in economic importance (Heredia-R et al., 2020). Oil extraction, an extremely charged issue in the Ecuadorian Amazon, drives both market integration and breakdown of traditional economic and cultural practices (Bozigar et al., 2016). With increasing market integration, migration to urban areas is increasing (Davis et al., 2017). Nevertheless, compared with other groups in the region, Kichwa tend to have lower levels of outmigration and longer durations of residence in or near their current dwellings (Vasco et al., 2018). People who depart ancestral villages for wage labor frequently return, further indicating constrained geographic mobility (Davis et al., 2017).

The lowland Kichwa continue to practice a form of small-scale agriculture known as *chagra*, which is characterized by extremely high levels of crop and product diversity on small plots attached to private residences (Heredia-R et al., 2020; Perreault, 2005). The *chagra* system, practiced for millennia, makes sustainable use of rainforest land that is poorly suited for more intensive agriculture (Heredia-R et al., 2020; Vera-Velez et al., 2019). Starting in the 1970s, many indigenous peoples began to ranch cattle, largely driven by a need to preserve land titles against claims by mestizo colonists and oil companies (Rudel et al., 2002). Nevertheless, *chagra* cultivation remains ubiquitous among the Kichwa (Heredia-R et al., 2020; Vera-Velez et al., 2019). Overall, then, Kichwa in Napo and nearby provinces have comparatively moderate levels

of market integration and stronger ties to particular localities than other ethnic groups in the region.

1.2.1. Kichwa Religion

Lowland Kichwa communities are largely Christian, with Protestantism and Catholicism varying in proportion across locales. However, the Kichwa approach to religion is syncretic, with traditional animistic beliefs and practices fluidly mingling with Christianity in daily life (Uzendoski, 2005; Uzendoski & Whitten, 2014). In line with Amazonian ethnic origins, traditional lowland Kichwa mythology and religion share more similarities with those of neighboring indigenous Amazonian groups than with other Quechua peoples (Grzech et al., 2019). Kichwa beliefs heavily focus on the life of the forest and on kinship among humans, as well as between humans and forest beings (Uzendoski, 2005). Complex, multipart wedding ceremonies are central events for the reinforcement and creation of these kin ties and help anchor a broader cosmology of kinship between humans and nonhuman spirits (Uzendoski & Whitten, 2014). A common theme in Kichwa religion is the interchangeability of human and forest spiritual essence, and indeed transformation in general (such as between plant and human "kin"; T. D. Swanson, 2009; Uzendoski, 2005). Shamans, called *yachaks*, are important (if sometimes morally ambivalent) figures in most communities and help navigate transformative relations between humans and spiritual powers, often in service of healing (Uzendoski, 2005; Uzendoski & Calapucha-Tapuy, 2012).

Plant products play crucial roles in the Kichwa cosmos. Ayahuasca, a potent hallucinogen, features in certain shamanic ceremonies (Naranjo, 1979; Salibová, 2020; Uzendoski, 2005). A spicy sauce made of *ají* peppers is used both in daily cuisine and for punishment of misbehaving children through administration to the eyes by shamans. Guayusa, a caffeinated tree leaf made into a tea, is particularly ubiquitous in daily and ceremonial life; an important morning ceremony, the *waysa upina*, revolves around the preparation and serving of guayusa tea by women (Suzina, 2023). *Chicha*, a fermented beverage typically made with yuca (manioc) or the savory tree fruit *chonta*, is also a symbolically potent presence at virtually every gathering (Tobes et al., 2022). *Mingas*, or work groups, are important focal points for Kichwa communities (Vasco et al., 2018), and participants can typically expect to be served *guayusa* and *chicha* by women at every *minga. Chagra* gardens are used to grow these products. Strong symbolic links thus exist between women, gardens, and the central consumables in Kichwa ritual.

Over the course of colonization and modernity, these links and the centrality for Kichwa life of the rainforest and its products (the *Kawsak Sacha* [living forest]) have become assimilated to a pan-Quechua concept and deity, *Pachamama* (Minoia et al., 2024). This name, originally of highland Andean origin, roughly translates to "the Space-Time Mother" or, perhaps less evocatively, "the World Mother" or "Mother Earth" (Inquilla-Mamani & Chambi-Apaza, 2019;

Kohn, 2013; Uzendoski & Calapucha-Tapuy, 2012). It is possible that deepening Pachamama beliefs have been incentivized by contact with outsiders, particularly sympathetic Westerners who associate Pachamama with environmentalism and ecology. Indeed, the Ecuadorian constitution of 2008 guarantees the rights of "Pachamama" (essentially, the natural environment), demonstrating a certain amount of cooptation by mestizo culture (Oikonomakis, 2024). Accordingly, our interviews revealed abundant conceptual links between Pachamama and ideals of protecting the environment, not polluting, and not cutting down trees. However, despite being an Andean deity grafted onto earlier layers of rainforest-focused spiritual beliefs, Pachamama has become a genuinely important personage in lowland Kichwa cosmology. Roughly speaking, she summarizes or stands in for the entire web of relations that make up the *Kawsak Sacha*, the people, and the spirits and powers that link them. For most lowland Kichwa in our sample, Pachamama therefore coexists unproblematically alongside the Christian God, the Virgin Mary, and countless other spiritual powers as a feminine personification of growth and land.

1.2.2. Yawari Village, Napo Province, Ecuador

Our field site was Yawari, a small (population ≈ 200) Kichwa village in the Rukullakta pueblo of Napo Province. Napo is the most populous province in the vast lowland Amazonian region east of the Ecuadorian Andes. Its Kichwa residents describe themselves as *Napo Runa* ("Napo people"; Uzendoski, 2005). Rukullakta is an autonomous Kichwa pueblo with around 2,300 inhabitants across 17 villages and more than 400 km^2, most of which is rainforest (Erazo, 2013; Varela, 2025). The pueblo sits a few kilometers northeast of the local canton town of Archidona and about half an hour's drive from the provincial capital of Tena. Rukullakta was formed in the late 20th century and incorporated in 2006 to assert local Kichwa land rights in the face of encroachment by mestizo colonists and mining companies (Erazo, 2013). Its charter mandate includes ecological protection of the rainforest as well as putting Kichwa land to sustainable economic use (Varela, 2025). Rukullakta villagers previously ranched cattle but now more commonly participate in cacao and coffee production to generate income, solidify land claims, and provide legal protection against oil mining on pueblo lands (Varela, 2025). Most households also maintain a *chagra.* A single-item question in our post-experiment interviews asked what percentage of food consumed in each participant's household the previous day was purchased at a store; the mean response was 51.1% (SD ≈ 30%), indicating only partial market integration, albeit with considerable inter-individual variation. Religiously, Yawari village is entirely, if often nominally, Catholic; there is no parish church, but a priest visits on Sundays to say Mass in the village community center.

The Napo region is a center of Kichwa culture, with around 56% of the province's residents claiming indigenous identity as of 2003 (Informe Social, 2003). Triumphal statues of Jumandi, a local Quijo hero killed in an important 16th-century uprising against the Spanish, stand

prominently in traffic circles at the northern entrances to both Tena and Archidona. Relations between Kichwa and mestizo colonists are characterized by both prejudice and transactional pragmatism; many Kichwa live in the urban centers and work in professional jobs, and local mestizos have frequently assimilated to Kichwa customs such as drinking guayusa and eating *chontacuro* palm weevil larvae (Chimbo-Gándara et al., 2024). Nevertheless, cultural and class barriers ensure that Napo indigenous and mestizo identities remain separate. In a telling example, an oratory shrine for the recently canonized Catholic Venezuelan saint José Gregorio Hernández stands in a small jungle clearing just outside the main entrance to Yawari on the road to Archidona. Despite being less than 1,000 feet from the Yawari Community Center, the shrine attracts almost exclusively mestizo worshippers, and Kichwa informants in Yawari generally disdain the saint and his mestizo devotees.

Compared with non–pueblo-dwelling Kichwa, then, Rukullakta Kichwa may be particularly culturally insulated, with many residents only occasionally venturing to Tena or interacting with outsiders. Pueblo inhabitants take their orientation instead from the opposite direction, the forest and its symbolic history. In particular, the isolated mountain *Yawar Urku* (mountain of blood) rises from the forested lowlands near Rukullakta. The mountain was the site of a successful battle against the Spanish during the colonial era (Erazo, 2013). Kichwa in the pueblo, and especially Yawari, the mountain's namesake, strongly identify with *Yawar Urku* and consider it an important sacred site, frequently assimilating it to the (separate) historical Jumandi uprising. Hence, the Yawari village name evokes both the local sacred mountain and the concept of "blood," with resonances both of intergenerational kinship and historical struggle. Kichwa identity in our field site is thus strongly oriented around symbolic touchstones of rainforest life and forceful assertions of Kichwa identity against encroaching outsiders.

1.2.3. Ethnographic Interviews

Our experimental and ethnographic methods required that we select two representative, culturally salient deities — one "moralistic" and one "local" — about which to ask questions and to ultimately examine the role of "local" beliefs on intergroup cooperation. To represent "moralistic" traditions, we needed to select a "moralistic" god as our target. We anticipated that the Christian God would serve as the moralistic deity in Yawari, but in the absence of extant research, it was unclear whether belief in Pachamama would be sufficiently salient for an adequate local god. It was also possible that Catholic saints, local Marian apparitions (which are plentiful), or more specific local or forest spirits would better serve in that role. We therefore conducted preliminary ethnographic interviews with Yawari residents (n = 20) to gather both quantitative and free-list data about local religious beliefs. These Religious Landscape Interviews — so called because they represent the first-sweep effort to understand local spiritual outlooks — were conducted in Spanish, as were all subsequent experiments and games.

Free-list interviews allow interviewers to gather naturalistic cultural knowledge that can be used to calculate cultural salience (Purzycki, 2025a). For example, the first item in our preliminary interviews asked subjects to simply "Please list all of the gods or spirits that you can think of that people believe in here [Yawari]." Subjects then ranked all the gods or spirits they listed in order of importance, then indicated whether each one (a) was concerned about how people behave; (b) could see into human beings' thoughts and feelings (i.e., was psychologically omniscient); (c) punished people for bad behavior; and (d) rewarded people for good behavior. Salience for each deity was calculated using Smith's *S* (Smith & Borgatti, 1997):

$$S = \frac{\Sigma(\frac{n + 1 - k_i}{n})}{N}$$

Here, k_i is the item order number, n is the total length of an individual's list, and N is the sample size for the interview. Thus, if every single respondent listed the same god and listed it first, *S* would = 1.0 for that deity. To generate uncertainty intervals for these salience estimates, each deity listed by subjects was modeled independently using an ordered beta-distributed model with uninformative priors (Major-Smith & Purzycki, 2026). Analyses were conducted in the AnthroTools package in R (Purzycki & Jamieson-Lane, 2017). Respondents listed 16 deities in total, but only the Christian God (n = 15), Pachamama (n = 15), Jesus Christ (in various formulations; n = 7), and the Virgin Mary (in various formulations; n = 6) were listed by more than five respondents.[2] Salience analysis indicated that the Christian God and Pachamama were by far the most culturally salient deities (Figure 1). Analyses of follow-up items indicated that respondents viewed the Christian God as somewhat more punishing, omniscient, and likely to reward good behavior than Pachamama, though both deities were equally concerned about morality (Table 1). As such, we selected the Christian God as the "moralistic" god and Pachamama as the "local" god for the experiments and interviews.

[2] Pachamama has often been identified with the Virgin Mary in Quechua and Andean contexts elsewhere in South America (Damian, 1995), raising the question whether the two deities would be better treated as one overlapping category in our dataset. However, we found no evidence for this assimilation, and in fact numerous Yawari respondents informally — but forcefully — asserted that Pachamama was *not* the Virgin Mary, indeed seeming puzzled and possibly offended by the question. We concluded that Pachamama was validly treated as a distinct deity in our sample.

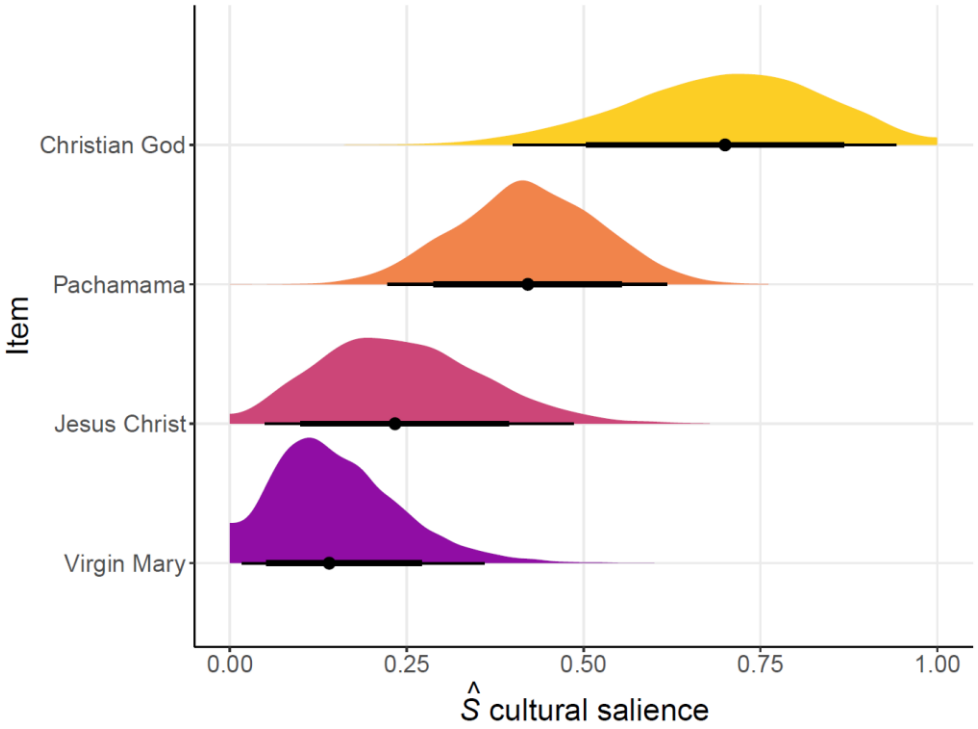


Figure 1. Cultural models of deities. Black circles are Smith's *S* values, with error bars displaying 89% (thick lines) and 95% (thin lines) credible intervals. Deities listed by 5 or more of the 20 Religious Landscape Interview participants are included here.

Table 1: Punishment, knowledge breadth, moral concern and reward of most salient deities, based on Religious Landscape Interviews (*n* = 20).

	Punishes behaviour	**Sees thoughts and feelings**	**Morally concerned**	**Rewards behaviour**
Christian God	12/15 (80.0%)	15/15 (100%)	15/15 (100%)	15/15 (100%)
Pachamama	11/15 (73.3%)	10/15 (66.7%)	15/15 (100%)	14/15 (93.3%)
Jesus Christ	4/7 (57.1%)	6/7 (85.7%)	7/7 (100%)	7/7 (100%)
Virgin Mary	2/6 (33.3%)	3/4 (75.0%)	5/6 (83.3%)	5/6 (83.3%)

2. Methods

2.1. Theoretical Model

Our theoretical causal model is illustrated in Figure 2 (this model was adapted from our core causal model for the wider Gods, Games, and the Socioecological Landscape project, available here). Our focal estimand is the causal impact of beliefs in supernatural punishment and omniscience of the local deity (Pachamama) on impartial fairness and generosity towards distant co-religionists of the same local religious tradition (i.e., fellow Kichwa believers in Pachamama). We believe the following sociodemographic variables may potentially cause both Pachamama beliefs and extended religious cooperation (Albrecht & Heaton, 1984; Baimel et al., 2022; Purzycki & Bendixen, 2024; Scheepers et al., 2002; Vardy et al., 2022) and thus treat them as confounders of this effect: age, sex, years of formal education, current food security, and number of children. We also include "market integration" as an additional confounder. That is, Kichwa who are more reliant on *chagras* and hunted or foraged food may both have a stronger connection with the land than market-integrated neighbors, and hence stronger Pachamama beliefs, as well as perhaps a greater need for more expansive social and cooperative networks. Alternatively, more market-integrated individuals may be more aware of an abstract identification with anonymous Kichwa in a neighbouring province and connect that identification with Pachamama, whereas subsistence farmers may be more focused on their own locality and thus see distant Kichwa as less salient. Either way, market integration is a potential confounding factor.

Further, while game order and comprehension checks are not confounders (they cannot cause our exposure), we included them to minimize uncertainty around the focal estimand, given that they likely explain variation in cooperative behaviour. Because participants are assumed to not regularly interact with distant co-religionists, we assume that participants' beliefs — which are the end result of years of enculturation and individual processes — are not the result of their behavior in these games, minimising the risk of reverse causality bias. Note, however, that we cannot rule out prior cooperative behaviour outside these games causing both subsequent religious beliefs and cooperative game behaviour. We must also assume that there is no unmeasured confounding (denoted by the node '*U*' in Figure 2), measurement error, or selection/collider bias in order to satisfy the requirements of causal inference (Hernán & Robins, 2020).

In addition to the core confounders introduced above, other variables may plausibly confound our target estimand. These include Catholic religious beliefs and rituals and engagement in Pachamama rituals. While each of these variables may plausibly cause cooperation with distant Pachamama co-religionists, given the cross-sectional nature of the data it is impossible to know whether these variables cause Pachamama punishment and omniscience beliefs (i.e., they are

confounders, and need to be adjusted for to remove bias), or whether they are *caused by* Pachamama beliefs (i.e., they are mediators of our focal estimand, and do *not* need to be adjusted for when estimating our causal effect of interest). Causal effects in both directions are also possible; hence the bidirectional arrows between these variables in Figure 2. For instance, stronger beliefs regarding Catholicism and participation in Catholic rituals might result in weaker beliefs regarding Pachamama; equally, stronger beliefs regarding Pachamama might result in weaker Catholic beliefs and practices. Or, perhaps beliefs in both stem from some common underlying latent or unmeasured factor, such as overall religiosity or other cognitive faculties. Similar bidirectional effects are also plausible for engagement in Pachamama rituals (i.e., it could cause, and be caused by, Pachamama beliefs). Given this ambiguity, using a single model to estimate an unbiased causal effect is impossible, and instead sensitivity analyses of multiple statistical models corresponding to different causal data-generating mechanisms are required (Major-Smith, 2023). These will be discussed in more detail below.

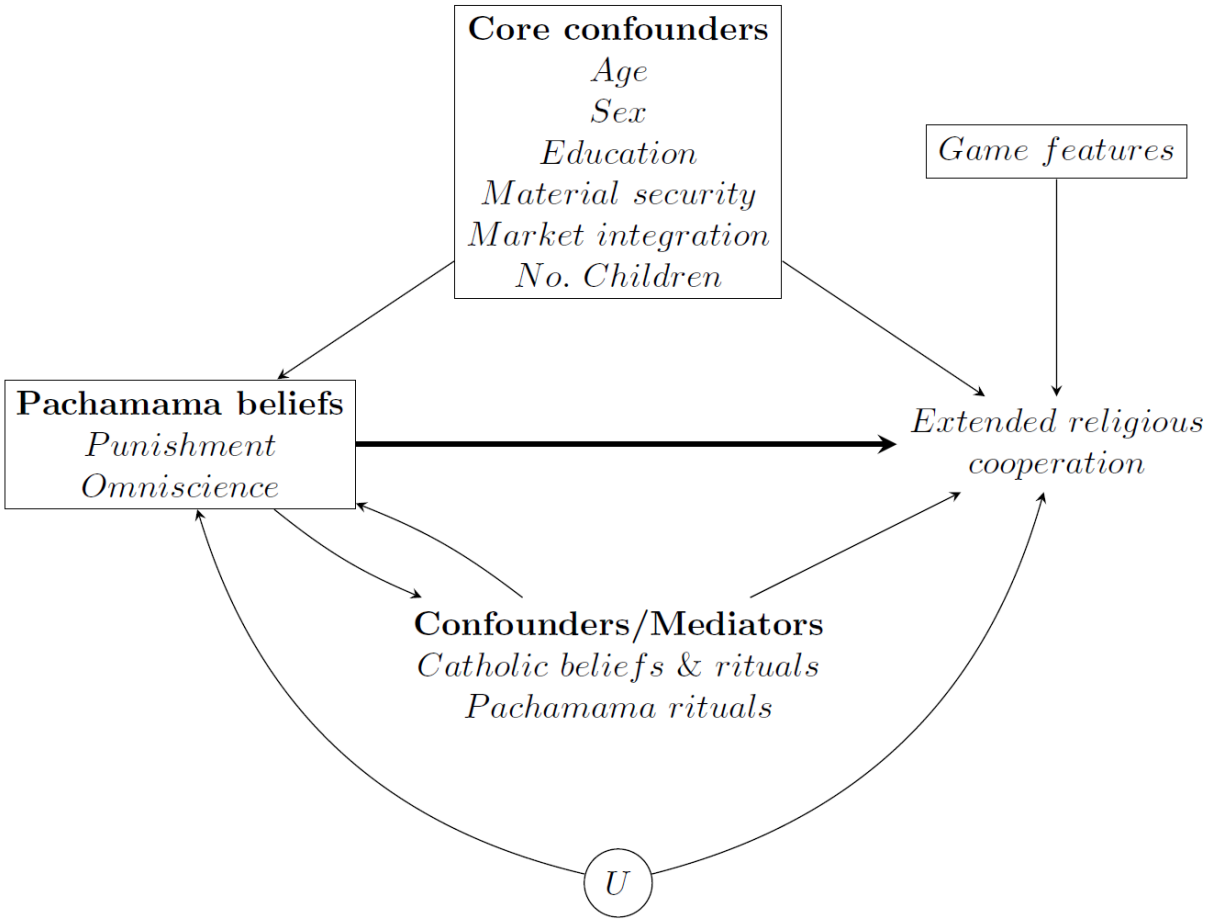


Figure 2. Causal graph displaying our core causal assumptions regarding the data-generating process.

2.2. Materials and Procedure

2.2.1. Demographics

We asked a battery of standard demographic questions. Most relevant to this project are the core confounding variables of age, sex, years of formal education, current food security, number of children, and market integration discussed above. For years of formal education, we simply asked for all years of education in a formal school setting, and for food security, we asked whether or not participants worried about getting enough food to eat over different time horizons (month/6 months/year/5 years; yes = 1; these four questions will be averaged together). For market integration, participants were asked the proportion of food purchased at a store, both for the preceding day and the past month.

2.2.2. Beliefs

Our focal exposures are supernatural punishment and omniscience of Pachamama, which were measured as part of a larger religiosity survey, separate from the Religious Landscape Interview, assessing beliefs and practices towards both the Christian Catholic God (our "moralistic" god) and Pachamama (the "local" god). We measured punishment using the mean of two questions: (1) Does **Pachamama** ever punish people for their behavior? (yes = 1, no = 0) and (2) Can **Pachamama** influence what happens to people after they die? (yes = 1, no = 0). To measure Pachamama's omniscience/knowledge breadth, we took the mean score of: (1) Can **Pachamama** see into people's hearts or know their thoughts and feelings? (yes = 1, no = 0) and (2) Can **Pachamama** see what people are doing if they are far away, in Guayaquil (Ecuador's largest city, 270 km away on the Pacific coast)? (yes = 1, no = 0).

The same questions as above were used to assess punishment and omniscience beliefs regarding the Christian Catholic God (with "Pachamama" replaced by "Dios" [God]). For Pachamama and Catholic rituals, we used the question "How often do you perform rituals or ceremonies devoted to **[deity]**?", answered on a five-point Likert scale (0) Very rarely/Never; 1) A few times per year; 2) A few times per month; 3) A few times per week; 4) Every day or multiple times per day).

2.2.3. Cooperation

To measure impartial fairness and generosity, we deployed variants of the Random Allocation Game (Hruschka et al., 2014) and the Dictator Game. In one variant — the Self Game — participants allocated money between themselves and an anonymous, geographically distant co-religionist from Palimbe, a randomly chosen Kichwa village in the neighboring province of Pastaza. Palimbe lies deep in the rainforest, a significant distance along poor roads from

Pastaza's provincial capital of Puyo. No one in Yawari had heard of Palimbe or claimed to know or be related to anyone who lived there. The other variant — the Local Game — entailed participants allocating money between a co-religionist from their own community and another anonymous co-religionist from Palimbe. "Co-religionists" here were explicitly identified as fellow Kichwa believers in Pachamama. Participants played the Random Allocation Games first (counterbalanced across participants) followed by the Dictator Games (also counterbalanced).

In the Random Allocation Game, participants were given the option of two individuals — denoted by two labelled but otherwise identical cups — and asked to think silently of the individual to whom they wished to allocate a coin. They then had to roll a fair die with an equal number of black and white faces. If the die came up black, they had to put the coin into the cup they thought of, but if the die came up white, they had to put the coin into the *opposite* cup. In each variant of the Random Allocation game, participants allocated 30 $1 coins in USD (the legal tender currency in Ecuador) for a total of USD $30. The mean distribution to distant co-religionists in perfectly fair play would therefore be $15 for each iteration of the Random Allocation Game, on average. For the Dictator games, participants were given ten $1 coins to divide between the two cups however they wished. Across two games, then, the total amounts were a maximum of $80 (approximately an average daily wage).

2.2.4. Game Features

As mentioned above, we also include game order and comprehension checks as factors that might affect the outcome, but we assume these have no bearing on beliefs and demographics. The game order was randomized across participants, but we anticipate that playing the Self games second, for example, might induce a little more self-interested behavior. Game comprehension checks were designed to screen out individuals who did not understand the game (and hence did not play these games), but even among those who passed, the number of comprehension checks needed may have impacted game behaviour.

2.2.5. Procedure

To minimize collusion, we conducted all experiments first and followed up with demographic and religiosity interviews over the course of the following week. Data collection was conducted on a single day in the local school, in the center of Yawari village. Subjects played games alone in temporary tarp booths in classrooms facing a wall. The school grounds were walled, with only one entrance and exit to the street. The game area was part of a long "L" of classrooms enclosing a central outdoor auditorium. The only way to move between classrooms was via an outdoor concrete walkway that connected to the auditorium/gym space. Participants gathered in the auditorium area to wait their turn to play, and barriers of tarp were set up to prevent visual access to the walkway and classrooms. We provided snacks, drinks, and lunch in the waiting area and,

throughout the day, called participants one at a time to exit the waiting area via an opening that faced away from the classrooms. They then were led via the walkway to a classroom booth, given an explanation of the games by a research assistant, tested for comprehension, and allowed to play. Trays of coins and games were prepared in a separate classroom and were shuttled out to the booths by research assistants, who also collected completed game trays for counting in the preparation room. Participants were given numbered ID cards with randomization codes for game order and were told to keep the cards for the interviews the subsequent week. Upon completing the games, participants were escorted to the school exit in the same direction as the flow of participants from the waiting area. All movement from the waiting area to game booths to exit was thus one-way. At the exit, participants signed out with a volunteer. Signing out entailed selecting an interview slot for the upcoming week. Participants were not allowed to reenter the school grounds after departing and were told not to talk about the game with anyone. No further participants were allowed into the school grounds after game play began in mid-morning. Participants in the waiting area were told not to discuss the game before playing. Phone use was not allowed in the waiting area.

One participant, a day laborer who would be unavailable the following week, carried out the demographics and post-game interviews on the day of game data collection in a classroom after other players had departed. All other interviews were conducted in daylong blocks over the ensuing week in the neighboring community center. Participants arrived on the day designated by their signup slot and were interviewed in different parts of the community center, with between four and five assistants conducting interviews in parallel. Food and drinks were provided. After completing interviews, subjects were paid the flat rate for their participation (Section 2.4) alongside their self-allocations from the Random Allocation and Dictator Self games, then signed out.[3] After all interviews were completed, a different group of Yawari and nearby Rukullakta residents was assembled in the community center for payouts for the Random Allocation and Dictator Local games. Each player's allocations were distributed to only one recipient, as indicated in the game instructions. However, multiple allocations were combined for each recipient to achieve a rough parity in distributions. At the end of the week, we traveled to Palimbe village, three hours' drive away, for randomized distribution of "distant" game allocations. Again, each player's allocations were distributed to only one recipient in Palimbe, but recipients in Palimbe all received allocations from several Yawari participants to even out distribution values.

[3] It should be noted that, for a small number of participants (8/86), allocations added up to irregular values (totals for one or more Random Allocation Games ≠ \$30 and/or Dictator Games ≠ \$10). This could indicate cheating or stealing on the part of game players, or human error on the part of experimenters loading trays, or both. Participants were nevertheless given the allocations they had distributed to cups in each game. We removed these values prior to analyses and in turn imputed them alongside other missing data for the main multiple imputation analyses (see below).

2.3. Statistical Model

As detailed in our causal graph (Figure 2), our primary exposure was the joint effect of beliefs about Pachamama's proclivity for punishment and omniscience/knowledge breadth, and the outcomes were coin allocations to the distant co-religionist across two variants of the Random Allocation and Dictator Games. In all statistical models, we adjusted for all core confounders of age, sex, education, food security, number of children, and material security to block any potential back-door biasing paths through these variables. Although not confounders, all models also held game order and comprehension checks constant (the latter for RAG models only, as there were no comprehension checks for the DGs) to improve precision in our estimates.

Beyond this base model, as discussed above, Catholic beliefs and practices and Pachamama ritual practices may or may not also be confounders or mediators of the focal estimand. If these variables *are* confounders, then they need to be adjusted for in regression models to estimate our estimand without bias; if they are mediators, then they do *not* need to be adjusted for, as adjustment would induce bias; if there is reciprocal causation — meaning these variables are both confounders and mediators — then both adjusting and not adjusting for these variables will lead to bias. Given this uncertainty, we compared models with all possible combinations of these variables as covariates in statistical models (i.e., four possible model combinations). If our estimate does not change based on these different assumptions, we can have more certainty that different causal assumptions regarding these variables do not impact our results. Alternatively, if our estimate is variable according to different assumptions, this uncertainty will be discussed when interpreting results. At the very least, our causal effect of interest likely falls between the estimates of our minimally and maximally adjusted models (risk of under-adjustment vs. overadjustment biases, respectively; Major-Smith, 2023).

We used aggregated logistic regression models for the Random Allocation Games and ordinal models for Dictator Game allocations. Analyses were run in Stan (Stan Development Team, 2026) using the "brms" package (Bürkner, 2017) for R version 4.4.1 (R Core Team, 2024) using four chains with 2,000 iterations (1,000 chains as warm-up) and weakly regularising priors. To ensure that chains were well-mixed and -powered, we visually assessed convergence between chains, r-hat values (ideally 1.00), and effective sample sizes.

To estimate the marginal causal effect of supernatural punishment and omniscience beliefs on cooperation, we used *g*-computation methods (Hernán & Robins, 2020). As we do not know the causal relations between punishment and omniscience with any certainty — and hence cannot estimate the independent causal effects of these variables — we estimated the *joint* causal effect of these variables together; i.e., the marginal change in the outcome on the count scale from punishment and omniscience both as "0" to both as "1," based on posterior predictions generated from our statistical models and the data.

We initially treated beliefs in an additive fashion, followed by an interaction effect between punishment and omniscience. We assessed which model was a better fit using LOOIC (see Bendixen et al., 2023), although due to a relatively small sample size (just over 80), our analyses may be underpowered to detect said interaction effect (Blake & Gangestad, 2020).

In the pre-registration document we only proposed conducting complete-case analyses, due to low levels of missing data in individual variables. However, after preparing data for analyses, the sample size of our complete-case data dropped more than anticipated, by approximately 30% (patterns of missing data are described in more detail below). To boost statistical efficiency and reduce the threat of selection bias due to missing data, we used multiple imputation by chained equations to impute this missing data (van Buuren, 2018; White et al., 2011). We imputed up to the 85 participants who provided any data for the demographics interview, religiosity interview and game sessions (86 participated in total, but one person was excluded here as they did not take part in the religiosity interview). We imputed 20 datasets, with a burn-in period of 10 iterations per imputation. The derived punishment and omniscience mini-scale variables were imputed via passive imputation. We used "participant believes in [deity (either Pachamama or Christian God)]" and "frequency participant thinks about [deity (either Pachamama or Christian God)]" as auxiliary variables in our imputation models to help predict missing values, particularly regarding the religiosity variables; however, as all participants believed in Christian God, this was not included in the imputation models. Given the observed patterns of missing data – small amounts of missingness across a number of variables – and our use of auxiliary variables, we believe our imputations meet the "Missing-At-Random" assumption required for unbiased imputations. Imputed datasets were analysed using the "brm_multiple" command in "brms" to run models on each imputed dataset, with LOOIC statistics for the imputed data averaged across imputed datasets. These imputed results form our main analyses, with complete-case analyses results also presented for comparison. Example scripts demonstrating the pre-registered analyses on simulated data are available here.

2.4. Sampling and Participants

Each participant was given $20 as a show-up fee. Participants were recruited via word of mouth. The small size of Yawari and abutting Rukullakta settlements (particularly Rukullakta village, another of the 17 constituent villages of Rukullakta pueblo, which is geographically contiguous with Yawari) allowed word to spread quickly. The sample thus recruited was somewhat disproportionately female (73%). It is possible that the female bias in our sample reflects a higher likelihood for men to work outside the village, but that is speculative. Another equally possible explanation is that men in Yawari are simply less likely to participate in social events than women. The mean age of participants was 41.4 (SD = 17.3), indicating strong age diversity.

As indicated in Section 1.2.2, participants reported growing about half their own food on average, suggesting limited market integration overall.

2.5. Pre-registration

A pre-registration for this paper can be found here. As data for this study had already been collected and audited, this was a pre-registration of our analysis plan, rather than for data collection and analysis. Plans for data collection can be found in the grant proposal, available here. This knowledge of the data and ethnographic context was useful to help understand some of the complexities of the data and help inform analysis decisions (e.g., the potentially complex relationship between the Christian God and Pacahamama, individual variation in market integration, and irregular game allocations). As also noted above, we initially developed core causal and statistical models for these site-specific analyses (here), which we then elaborated upon with the local context in mind for each of the site-specific papers. For full transparency, we note that descriptive statistics for key variables and regressions using this wider project core causal model were conducted in January 2026 for a project workshop, but the preliminary results of this analysis did not influence our causal model or corresponding statistical approach outlined in the pre-registration (posted May 2026).

Our analyses followed the pre-registration as specified, although due to greater-than-anticipated levels of missing data in our complete-case analyses we performed multiple imputation to impute these missing data, as described above.

3. Results

3.1. Descriptive Statistics

3.1.1. Sociodemographic Variables, Religious Beliefs, and Missing Data

Table 2 summarises the sociodemographic and religiosity data, for both raw and imputed data, as well as proportions of missing data for each variable. Across all variables with missing data, there is little difference between the raw and imputed results, although Pachamama beliefs are marginally lower in the imputed data, potentially suggesting that individuals with missing data for these questions were less likely to attribute these powers to Pachamama.

Consistent with the Religious Landscape Interview above, approximately three-quarters of participants believed that Pachamama punishes people for their behaviour, while around 60% believed Pachamama could influence what happens to people after they die. There was a modest positive correlation between these Pachamama beliefs (tetrachoric correlation: raw = 0.37; imputed = 0.39). More participants believed that the Christian God punished people (~82%) and influenced people in the afterlife (~88%), although the correlation between these beliefs was low (tetrachoric correlation: raw = 0.04; imputed = 0.00). The correlations between Pachamama and Christian God punishment (tetrachoric correlation: raw = 0.31; imputed = 0.27) and afterlife (tetrachoric correlation: raw = 0.45; imputed = 0.46) beliefs were moderate. The mean of the punishment mini-scale was correspondingly higher for Christian God (raw = 0.85; imputed = 0.84) compared to Pachamama (raw = 0.67; imputed = 0.66), with a moderate positive correlation between these mini-scales (polychoric correlation: raw = 0.50; imputed = 0.44).

Turning to knowledge breadth, approximately 70% of participants believed that Pachamama could know people's thoughts and feelings, with just over half believing that Pachamama could see what people are doing far away. These beliefs were much higher among Christian God, with over 95% of participants responding in the affirmative. These responses were highly correlated, for both Pachamama (tetrachoric correlation: raw = 0.91; imputed = 0.88) and Christian God (tetrachoric correlation = 0.77). The correlations between Pachamama and Christian God for "knowing thoughts/feelings" was low (tetrachoric correlation: raw = 0.06; imputed = 0.02), but moderate for "seeing far away" (tetrachoric correlation: raw = 0.51; imputed = 0.54). The mean of the omniscience mini-scale was substantially larger for Christian God (0.97), compared to Pachamama (raw = 0.64; imputed = 0.62), with a moderate positive correlation between these mini-scales (polychoric correlation: raw = 0.48; imputed = 0.52). Moderate positive correlations were also observed between the punishment and omniscience mini-scales for both Pachamama (polychoric correlation: raw = 0.62; imputed = 0.63) and Christian God (polychoric correlation: raw = 0.47; imputed = 0.46).

Together, these results match those of the ethnographic interviews, demonstrating that Pachamama is a less punitive and more locally bound deity compared with the Christian God.

Table 2: Summary of Kichwa sociodemographics and religious beliefs regarding punishment and omniscience towards both Pachamama (local god) and Christian God (moralistic god; *n* = 85). Proportions of missing data for each variable are also presented, along with mean values across the 20 imputed datasets (for variables with any missing data). Note that the cell counts for imputed data can include decimal places, as they are averages.

Variable	N	Raw data - N (%) or median (IQR)	Imputed data - N (%) or median (IQR)	Missing data - N (%)
Age (years)	85	32 (25 to 49)	NA	0 (0.0%)
Sex (male)	85	22 (25.9%)	NA	0 (0.0%)
Education (years)	83	12 (11.75 to 12)	12 (11.5 to 12)	2 (2.4%)
Number of children	85	2 (1 to 4)	NA	0 (0.0%)
Material security	85	1 (0.75 to 1)	NA	0 (0.0%)
Market integration 1 (% food from market yesterday)	82	50 (30 to 80)	50 (26.2 to 80)	3 (3.5%)
Market integration 2 (% food from market in past month)	76	50 (30 to 80)	50 (30 to 77.5)	9 (10.6%)
Pachamama ever punishes behaviour	83	63 (75.9%)	63.6 (74.9%)	2 (2.4%)
Pachamama influences people after death	81	48 (59.3%)	49.0 (57.6%)	4 (4.7%)
Pachamama knows thoughts and feelings	80	58 (72.5%)	58.8 (69.2%)	5 (5.9%)
Pachamama can see people far away	83	46 (55.4%)	46.0 (54.2%)	2 (2.4%)
Christian God ever punishes behaviour	84	69 (82.1%)	69.5 (81.8%)	1 (1.2%)
Christian God influences people after death	83	73 (88.0%)	74.0 (87.1%)	2 (2.4%)
Christian God knows thoughts and feelings	85	84 (98.8%)	NA	0 (0.0%)
Christian God can see people far away	85	81 (95.3%)	NA	0 (0.0%)

3.1.2. RAGs and DGs

As noted above, there were some irregularities in game allocations, resulting in small numbers of participants having missing game data (Local RAG = 6 missing [7.1%]; Self RAG = 4 missing [4.7%]; Local DG = 1 missing [1.2%]; Self DG = 1 missing [1.2%]). Participants in the local RAG gave 14.0 tokens on average (SD = 2.77) to distant Pachamama co-religionists, out of 30 (imputed data: mean = 14.0; SD = 2.75). This was comparable in the self RAG condition (raw data: mean = 14.2, SD = 2.76; imputed data; mean = 14.2; SD = 2.77). Figure 3 presents these results, relative to a hypothetical binomial distribution of fair allocations. Allocations in both RAGs were therefore biased against distant co-religionists, as formally confirmed when testing against a null binomial distribution using intercept-only Bayesian regression models (local RAG: odds ratio = 0.88, 95% credible intervals [CI] = 0.81 to 0.95; self RAG: odds ratio = 0.90, 95% CI = 0.83 to 0.98). Using a repeated-measures model, there was no evidence for a difference in allocations between the local and self conditions (difference on count scale = 0.21, 95% CI = -0.79 to 1.26).

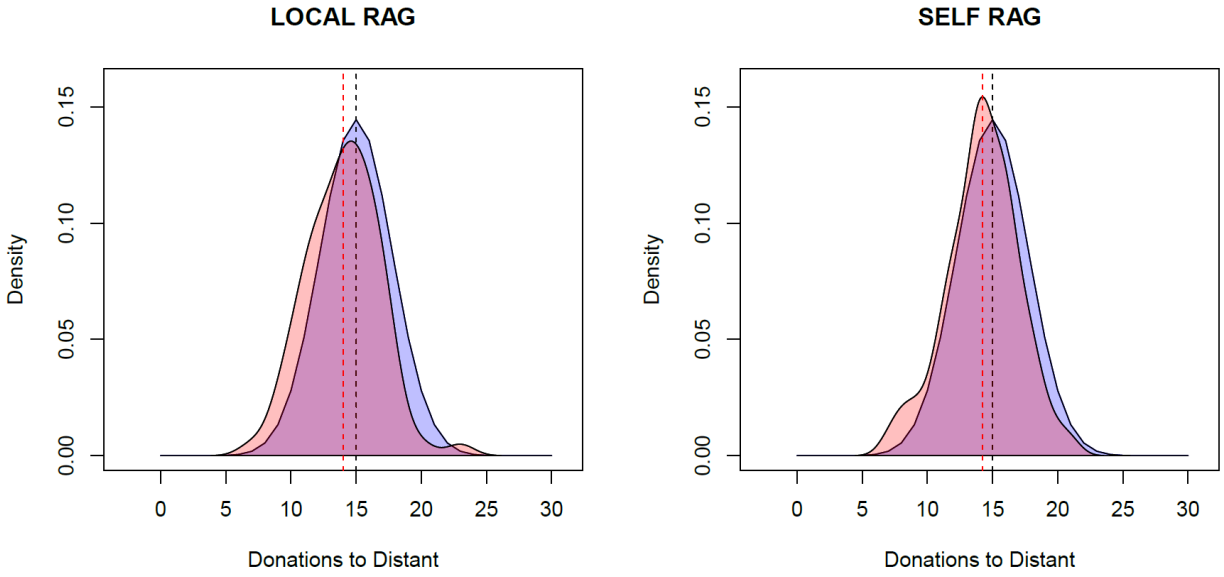


Figure 3: Random Allocation Game (RAG) donations to distant co-religionists in both the local (left) and self (right) conditions. The observed distribution is shaded in red, with the mean observed values indicated by the dashed red line; the theoretical binomial distribution assuming unbiased allocations is shaded in blue, with the mean expected theoretical value indicated by the dashed black line at 15 donations.

Dictator game allocations are shown in Figure 4. On average, participants gave 4.2 coins (SD = 1.90) to distant co-religionists in the local condition (imputed data: mean = 4.1; SD = 1.90), with a modal value of sharing five allocations (selected by 45.2% of participants). Donations were comparable in the self DG, with an average of 4.3 coins (SD = 2.24) given to distant co-religionists (imputed data: mean = 4.3; SD = 2.25); five coins was again the modal choice

(35.7%), albeit less pronounced than in the local DG condition. There was no evidence for a difference in donations between conditions in the DG using a repeated-measures model (difference on count scale = 0.02, 95% CI = -0.64 to 0.69).

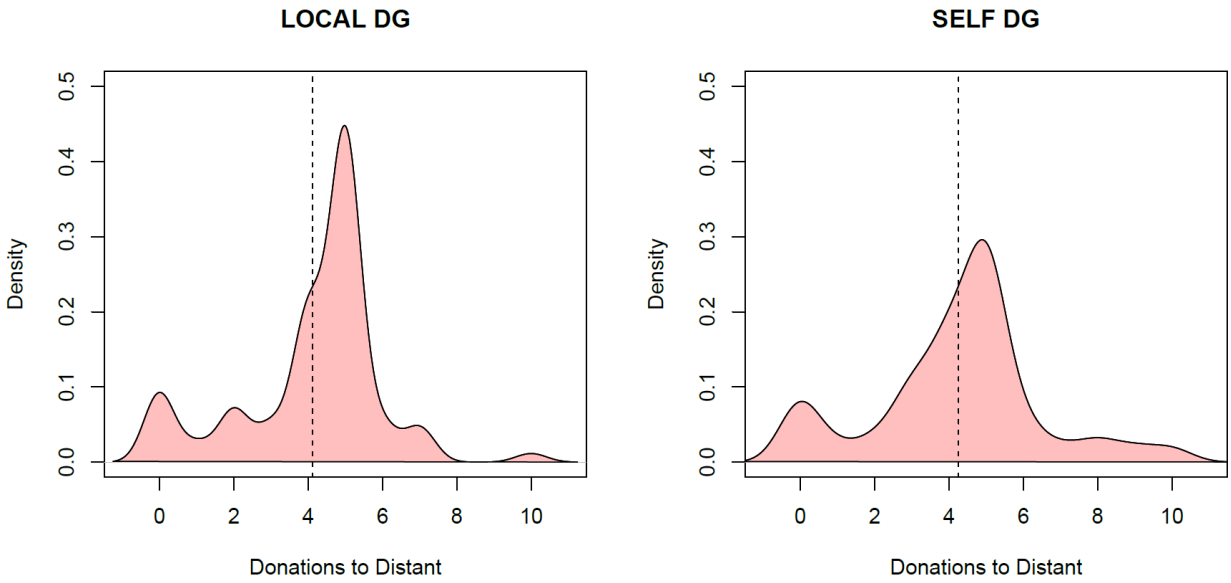


Figure 4: Dictator Game (DG) donations to distant co-religionists in both the local (left) and self (right) conditions. The observed distribution is shaded in red, with the mean observed values indicated by the dashed line.

3.2. RAG and DG Results

In this section, we present the results of our main regression and *g*-computation analyses, based on the causal assumptions outlined in the DAG above (Figure 2). Main analyses are from the multiply imputed data (n = 85), with complete-case results presented as a comparison (i.e., participants with complete exposure, outcome and covariate data across all model specifications; n = 58; 68.2% missing). A summary of all results, across all model specifications, is presented in Table 3.

3.2.1. RAG Results

We first present results for the "core confounders" in Figure 2 (i.e., age, sex, education, number of children, material security and market integration). For the local DAG, inclusion of an interaction term between Pachamama punishment and omniscience beliefs did not improve model fit (ELPD LOO for additive model = -210.6; ELPD LOO for interaction model = -211.5; ELPD diff = -1.0, SE diff = 0.3); we therefore performed our *g*-computation analyses based on the additive model. There was weak evidence for a negative relationship between Pachamama

beliefs and donations to distant co-religionists, with an increase in Pachamama punishment and omniscience beliefs associated with a one-unit decrease in donations on average, but with a range of plausible effect sizes also consistent with there being no relationship (median estimated difference in number of tokens to distant = -1.01, 95% CI = -2.99 to 0.96; from approximately 14.6 tokens when both exposures took the value "0" to 13.6 when both were "1"; Figure 5).

In the self RAG, there was also no evidence for a multiplicative effect of the exposure on allocations (ELPD LOO for additive model = -209.9; ELPD LOO for interaction model = -211.0; ELPD diff = -1.0, SE diff = 0.4). There was no evidence for a joint causal effect of Pachamama beliefs on donations (Figure 5; median diff = -0.07, 95% CI = -2.08 to 1.92; from approximately 14.2 tokens when both punishment and omniscience were "0" to 14.1 tokens when both were "1").

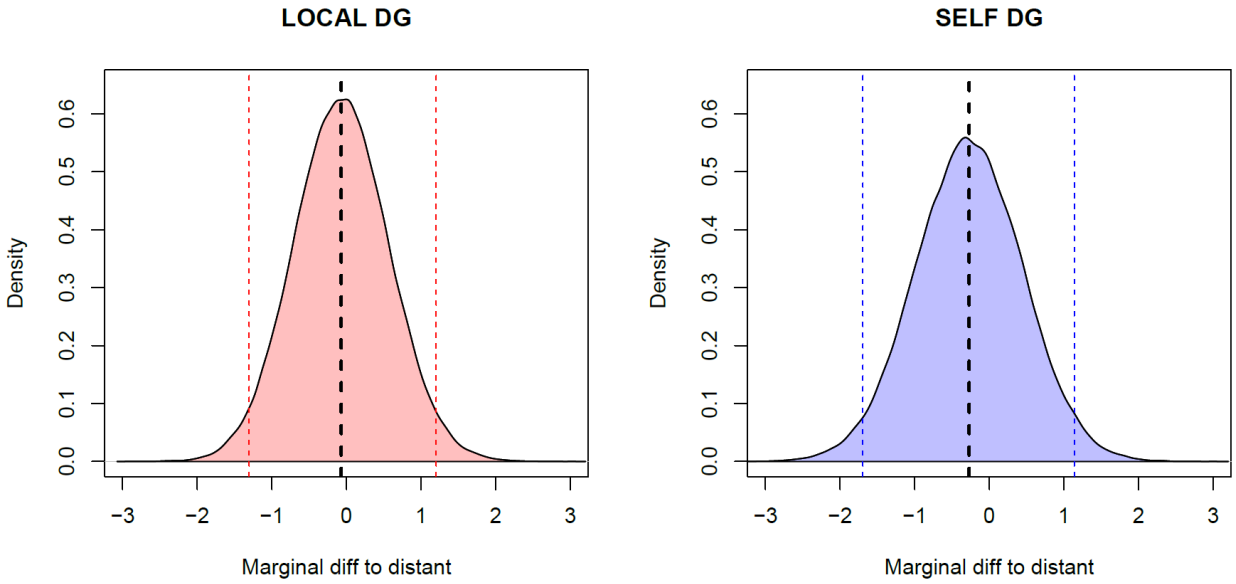


Figure 5: Marginal effect of change in Pachamama punishment and omniscience beliefs on change in cooperation with distant Pachamama co-religionists in the Random Allocation Games (RAGs; local RAG = left; self RAG = right). Dashed black lines denote the median expected difference in cooperation (number of tokens) from both punishment and omniscience taking the value "0" to both as "1," with 95% credible intervals in lighter dashed lines.

3.2.2. DG Results

Again focusing first on the model with core confounders, for the local DG an interaction term between exposures did not improve model fit (ELPD LOO for additive model = -156.2; ELPD LOO for interaction model = -156.5; ELPD diff = -0.3, SE diff = 0.2). Based on this additive model, there was again no evidence for a relationship between Pachamama beliefs and

cooperation with distant co-religionists (Figure 6; median diff = -0.07, 95% CI = -1.31 to 1.20; from approximately 4.2 tokens when both punishment and omniscience were "0" to 4.1 tokens when both were "1").

For the self DG, there was again no improvement in model fit when assuming a non-additive relationship between the exposures (ELPD LOO for additive model = -178.3; ELPD LOO for interaction model = -178.6; ELPD diff = -0.3, SE diff = 0.2). As with the local DG results above, there was no evidence for a joint effect of Pachamama beliefs on cooperation with distant co-religionists (Figure 6; median diff = -0.27, 95% CI = -1.69 to 1.14; from approximately 4.5 tokens when both punishment and omniscience were "0" to 4.3 tokens when both were "1").

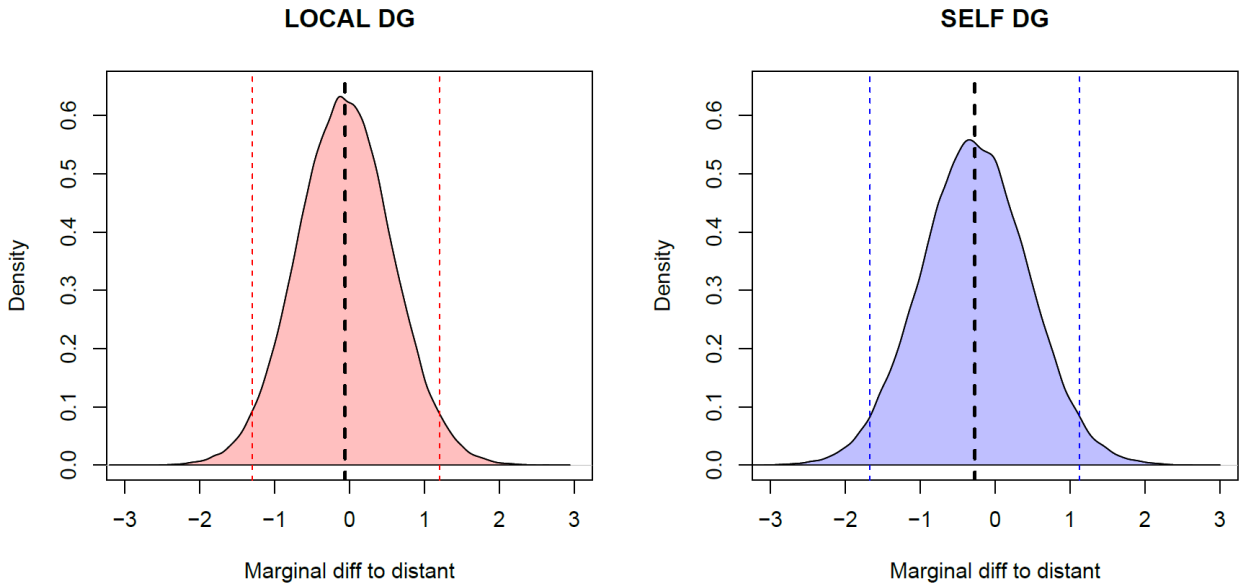


Figure 6: Marginal effect of change in Pachamama punishment and omniscience beliefs on change in cooperation with distant Pachamama co-religionists in the Dictator Games (DGs; local DG = left; self DG = right). Dashed black lines denote the median expected difference in cooperation (number of tokens) from both punishment and omniscience taking the value "0" to both as "1," with 95% credible intervals in lighter dashed lines.

3.2.3. Sensitivity Analyses and Complete-Case Results

Across all three sensitivity analyses examining the robustness of our results to different causal assumptions — i) including "Catholic beliefs and practices" as potential confounders; ii) including "Pachamama ritual practice" as a potential confounder; and iii) including "Catholic beliefs and practices" and "Pachamama ritual practice" as potential confounders — we observed little variation in the estimation of our focal contrast. That is, our results appear robust to different plausible causal assumptions regarding the structure of confounding. The complete-case

results are also broadly comparable to the imputed results, albeit with wider uncertainty intervals, as expected given the reduced sample size. This suggests that our complete-case results were not substantially impacted by selection bias, and/or that bias was not introduced during our imputation procedure. Either way, the similarity in results lends further support to the claim that our results are not unduly affected by selection bias.

Table 3: Summary of regression results. All estimates are the counter-factual marginal contrast in the number of allocations to distant Pachamama believers in both the Random Allocation Games (RAGs) and Dictator Games (DGs) comparing between when both punishment and omniscience beliefs regarding Pachamama take the values "0" vs. "1." These estimates are based on posterior predictions from the corresponding Bayesian model (either aggregate binomial for RAGs, or ordinal models for DGs). All models assume an additive relationship between punishment and omniscience on cooperation. Covariates in the main analysis include age, sex, education, number of children, material security, market integration and game features. Sensitivity analysis 1 includes Christian punishment and omniscience beliefs and practices as additional covariates. Sensitivity analysis 2 includes participation in Pachamama rituals as an additional covariate. Sensitivity analysis 3 includes both Christian beliefs and practices and Pachamama ritual participation as additional covariates. Point estimates are median values, with 95% credible intervals in brackets. Results are presented for both the multiple imputation results (MI; $n = 85$) and the complete-case analysis results (CCA; $n = 58$).

	Local RAG		**Self RAG**		**Local DG**		**Self DG**	
	MI	*CCA*	*MI*	*CCA*	*MI*	*CCA*	*MI*	*CCA*
Main analysis	-1.01 (-2.99 to 0.96)	-0.78 (-3.02 to 1.57)	-0.07 (-2.08 to 1.92)	0.09 (-2.28 to 2.41)	-0.07 (-1.31 to 1.20)	1.10 (-0.09 to 2.28)	-0.27 (-1.69 to 1.14)	-0.40 (-1.84 to 1.10)
Sensitivity analysis 1	-1.15 (-3.24 to 0.93)	-1.59 (-4.14 to 1.05)	0.06 (-2.00 to 2.11)	0.28 (-2.22 to 2.72)	-0.20 (-1.41 to 1.01)	0.97 (-0.21 to 2.16)	-0.20 (-1.62 to 1.21)	0.59 (-0.86 to 1.93)
Sensitivity analysis 2	-1.07 (-3.08 to 0.92)	-0.69 (-3.12 to 1.66)	-0.07 (-2.08 to 1.92)	0.02 (-2.48 to 2.34)	-0.06 (-1.28 to 1.19)	1.05 (-0.16 to 2.33)	-0.29 (-1.71 to 1.09)	0.53 (-0.88 to 1.95)
Sensitivity analysis 3	-1.19 (-3.28 to 0.89)	-1.41 (-3.98 to 1.16)	0.08 (-1.98 to 2.13)	0.22 (-2.29 to 2.59)	-0.20 (-1.41 to 1.01)	0.97 (-0.28 to 2.19)	-0.22 (-1.62 to 1.19)	0.53 (-0.83 to 1.90)

4. Discussion

Assuming that a causal interpretation is warranted (discussed below), these results indicate that, overall, Pachamama punishment and omniscience beliefs do not affect cooperation with distant Pachamama co-religionists among this Kichwa population. At best, beliefs in Pachamama's punishment and knowledge may promote in-group favoritism, as suggested by the weak estimated effect of the Local RAG game (although the range of estimates is also consistent with there being no relationship, and this result was not replicated in the Local DG game). These null findings were consistent across all other games, conditions, and sensitivity analyses and stand in contrast to previous cross-cultural work using these methods (Lang et al., 2019; Purzycki et al., 2016), which have found a small positive relationship between supernatural punishment beliefs and cooperation with distant co-religionists, albeit predominantly among "moralistic" religious traditions rather than "local" ones.

A number of possible reasons may explain these null results, none of which are mutually exclusive. First, the previously reported effect among moralistic traditions was quite small and only appeared in aggregate across multiple fieldsites; in many individual sites, the evidence for a relationship between supernatural punishment beliefs and cooperation within the moralistic religious tradition was much weaker and more inconsistent (Lang et al., 2024; Purzycki et al., 2024). Therefore, the comparison of this single Kichwa site with a local deity against a better-powered multi-site analysis with moralistic deities may not be valid. Additionally, the fact that the effect of moralistic deity beliefs on cooperation was not found across all previous societies suggests that any effect may be heterogeneous between populations; the same may be true of local religious traditions. Put another way, we simply cannot infer that local religious traditions are less effective *overall* at expanding the circle of cooperation than moralistic ones from this study, even if the effect of Pachamama beliefs on cooperation in this population is largely null.

However, preliminary results from other sites as part of this project do suggest that this null effect of local gods is found in other populations, tentatively supporting the interpretation that local god beliefs do not influence anonymous cooperation as moralistic god beliefs do (Ichinkhorloo et al., 2026; Mateffy et al., 2026; Norge et al., 2026). A more rigorous test of this hypothesis among the Kichwa specifically would explore the effects of both Pachamama (local) and Christian (moralistic) supernatural punishment beliefs on cooperation with distant Pachamama and Christian co-religionists (respectively), examining whether the effect of Christian beliefs on anonymous cooperation was stronger than that of Pachamama beliefs; unfortunately, given the constraints of data collection for this field site, such an investigation was not possible in the present study (but for a similar approach in a Benin population, see Dalla Ragione et al., in prep.).

Second, assuming that punishment and omniscience beliefs from local religious traditions *are* less effective at promoting cooperation with distant, unrelated, and unknown co-religionists than those of moralistic religious traditions, the question arises: why might this be so? While many have appealed to these gods' specific "moral concern" (or lack thereof; Baumard & Boyer, 2013; Norenzayan, 2013b; Norenzayan et al., 2016), cross-cultural work, again predominantly among moralising religious traditions, indicates that gods' perceived moral concern does not appear to drive cooperation (Bendixen et al., 2023). If moral concern does not explain any putative difference of supernatural punishment beliefs on cooperation between "moralistic" and "local" traditions, it is a little unclear what does, especially based on the cultural evolutionary theories noted above. One tentative explanation is that the current work only focused on binary questions of punishment and omniscience, perhaps obscuring important differences in the type, scale, frequency and severity of punishment; if these are greater in moralistic traditions, this may provide stronger motivation for individuals to behave cooperatively. This possibility requires further empirical testing.

Alternatively, perhaps local gods do motivate cooperation, just not the kind of cooperative behaviour measured by these games. For instance, as described above, Pachamama in Yawari mainly appears to care about how people interact with the environment, *chagra* gardens, the forest, and related domains, rather than interpersonal moral behaviour (especially impersonal cooperation with unknown and distant co-religionists using currency). Examining whether Pachamama beliefs or ritual practices measurably impact ecological behaviours — such as protecting the rainforest or moderating pollution and resource use — would be an interesting avenue for future research (see also Discussion in Norge et al., 2026). This possibility would neatly align with the theoretical expectation that religions have culturally evolved to solve a range of local socioecological challenges in which cooperation is often more broadly construed than simple economic fairness (Bendixen et al., 2024; Purzycki, Bendixen, et al., 2022).

For example, if local rituals and beliefs are adaptive for avoiding the tragedy of the commons in rainforest resource use, Pachamama belief in particular could serve as an "anchor" that stabilizes the broader cooperative equilibrium within the community by, say, preventing overharvesting. Pachamama beliefs may also be indexed to specifically Kichwa forms of local cooperation such as *minga* work groups. If so, beliefs that Pachamama is punitive and omniscient could plausibly be causally connected with *minga* participation but not monetary fairness. In Yawari and Rukullakta in particular, Pachamama may be more closely linked with protection of local Kichwa land against mestizo and foreign outsiders and companies, a major *raison d'être* of the pueblo. Failing to pull one's weight in the struggle to make the community productive enough to sustain indigenous land claims may be more salient for Pachamama punishment than cooperating with distant Kichwa.

Similarly, cosmological rituals such as weddings or even *waysa upina* ceremonies may plausibly both be associated with Pachamama and serve adaptive functions by reinforcing shared narratives that enable people to cooperate more easily in horticulture or food sharing within the local community. Impersonal generosity toward members of a far-off community, even if Kichwa, would in this case not necessarily be part of Pachamama's ambit. This possibility gains support from the intense relationship between Kichwa religion and kinship (Uzendoski, 2005), suggesting the possibility of a tight local boundary for moral concerns. Regardless, the present results suggest that current formulations of the supernatural punishment hypothesis — in which beliefs in punitive and omniscient deities uniformly motivate cooperative behaviour (Johnson, 2015; Johnson & Krüger, 2004) — are too simplistic and highlight future topics for empirical and theoretical development in this area.

Although the majority of results were consistent with a null effect, there was potentially weak evidence for a relationship between Pachamama beliefs and local favouritism in the Local RAG game. That is, the bulk of the posterior estimates suggested a negative relationship, such that Pachamama punishment and omniscience beliefs were associated with more tokens routed to local, rather than distant, co-religionists (Figure 5). If genuine, this effect could indicate a relationship between Pachamama and strictly local concerns, such as cooperative agricultural production or protecting shared pueblo land and local sacred places, which may favour the local in-group compared with distant co-religionists. This relationship may in turn reflect either: i) an overall effect in the population; or ii) effect heterogeneity, where the effects of Pachamama beliefs on Local RAG cooperation are stronger for some individuals than others. However, there was no similar effect in the Local DG (Table 3). It is not clear why Local RAG and DG results would differ if Pachamama promotes parochial, but not distant, cooperation. Furthermore, although most estimates for the Local RAG were negative, many were still positive, meaning the evidence for a negative relationship is quite weak and plausibly due to random variation (meaning the true causal effect may be null). The data here cannot adjudicate between these different possibilities, meaning that interpretation of this result is ambiguous.

4.1. Strengths and Limitations

This study boasts a number of important strengths, including: i) Combining a locally relevant and tailored ethnographic approach with state-of-the-art quantitative causal inference methods, to ensure that data collection and analyses are both ethnographically meaningful and analytically rigorous (Deffner et al., 2024; Purzycki & Major-Smith, 2026); ii) Use of formal causal inference methods with a clear estimand (i.e., our target estimate of interest; Lundberg et al., 2021) and clear details of the assumptions required for a causal interpretation from observational data (Hernán & Robins, 2020; Rohrer, 2018); and iii) A range of sensitivity/robustness checks — i.e., different assumptions regarding confounding, imputation methods for missing data and potential selection bias — that provide stronger evidence for a causal interpretation.

We also acknowledge a number of limitations, many of which revolve around our ability (or lack thereof) to infer causality from these data. For instance, from observational data there is always the risk of bias due to unmeasured confounding — i.e., variables that cause both Pachamama beliefs and cooperation, but which were not measured and/or adjusted for in our statistical model. Wealth and personality are two possible examples, although often the relationship of these factors to religiosity and cooperation is quite small and variable between populations (Henrich et al., 2015; Major-Smith, Morgan, Halstead, et al., 2023; Major-Smith, Morgan, Tohidinik, et al., 2023; Volk et al., 2011), potentially lowering the risk of this potential bias. In the context of Yawari, it is possible that high social status is associated with both strong commitment to Kichwa beliefs and values (including Pachamama) and a greater inclination to prefer to funnel resources into the local village rather than distributing them abroad. However, this is merely an example of a possible unmeasured confounder relationship, and one which is largely speculative given a lack of empirical data on these measures. As noted above, a range of robustness checks to different causal assumptions regarding confounding, as well as the imputation of missing data to lower the risk of selection bias, improve the case for a causal interpretation of our findings. Finally, for a causal interpretation to hold, we have to assume no measurement bias; this may be especially difficult to know for internal mental states such as religious belief, for which objective validation is incredibly difficult. This desideratum may be slightly complicated by the fact that Pachamama is originally an exogenous import in lowland Kichwa cosmology and is frequently coopted by outsiders for political or eco-activist ends (Oikonomakis, 2024). Reports about Kichwa belief may thus be conflated with a variety of political concepts and associations external to our target estimand. Nonetheless, our findings are in many ways compatible with previous ethnographic reports of Kichwa religiosity (Uzendoski, 2005; Uzendoski & Calapucha-Tapuy, 2012; Whitten & Whitten, 2008). These ethnographies emphasize many features found in our interviews, including cosmological links between femininity, gardening, and the forest; strong concepts of kinship across spiritual beings and humans; and the importance of marriage and ceremonies such as *waysa upina* as anchors for an expansive forest cosmology. These overlapping themes lend some support to the validity of our measurements for this study.

A further limitation is that of our relatively small sample size, which may increase the risk of type II errors (i.e., false negatives). Most point estimates were largely null, suggesting that any relationship may actually be negligible, but we cannot rule out small — though potentially meaningful — population-level effects (e.g., 95% credible intervals were generally +/- 2 tokens in the RAGs, quite a wide range). We did perform an a priori power analysis, which indicated that a sample size of 40 per condition could detect a 4%-point difference in RAG cooperation (i.e., approximately a one-token difference) with 90% power, many factors can impact statistical power (including variation in exposure and outcome, covariates, missing data, etc.), so our confidence in the power analysis may be tempered. Nevertheless, our findings are in line with

preliminary analyses for other field sites in this project (Ichinkhorloo et al., 2026; Mateffy et al., 2026; Norge et al., 2026), suggesting that even in a larger sample, little-to-no relationship would likely have been observed.

A further concern regards perceptions of Kichwa identity towards distant and unknown Pachamama co-religionists, a necessary pre-condition for a valid interpretation of our games as cooperation with a co-religious in-group. Kichwa settlements are widely distributed throughout culturally and geographically distinct regions of the Ecuadorian Amazon (Vasco et al., 2018). Lowland Kichwa ethnicity is thus often seen as somewhat more porous and diffuse than that of neighboring groups such as the Achuar or Huaorani, and Kichwa people are often considered more acculturated to Ecuadorian mestizo society in the Amazon than other indigenous groups (Whitten & Whitten, 2008). Conceivably, the comparative broadness of Kichwa ethnicity and its permeability *vis-à-vis* the mestizo majority may have dampened the salience of Kichwa identity among our Yawari subjects *vis-à-vis* distant inhabitants of Palimbe. In other words, it is possible that residents of Yawari did not see Palimbe recipients as genuine members of an extended co-religionist community, potentially muting any effects of Pachamama belief on cooperation. Our results should perhaps be cautiously interpreted in light of this possibility. Importantly, though, we encountered abundant evidence of fiercely salient Kichwa identity during our stay in Yawari, marked by verbal expressions of distrust of outsiders — particularly white Westerners, but also very much including Ecuadorian mestizos. (Recall, for instance, the dismissive attitudes of Yawari residents toward the nearby mestizo shrine of Saint José Gregorio Hernández.) When asked, many participants also indicated strong conceptual links between Palimbe inhabitants and Kichwa activities and rituals, particularly *waysa upina* ceremonies and *ají* punishments, as well as Catholic and Pachamama beliefs. All things considered, it therefore seems likely that a shared (and bounded) co-religionist identity was salient during the experiments and interviews.

4.2. Conclusion and Future Directions

We found little evidence that local Pachamama punishment and omniscience beliefs impact cooperation with distant Pachamama co-religionists among this Ecuadorian Kichwa population. This may suggest that local deities and religious traditions are less effective at promoting cooperation than moralistic ones, although there are many open questions for future work. Outstanding research agendas thus include: i) Comparing local and moralistic religious traditions in the same population; ii) Exploring the scope of local gods beliefs in promoting other forms of cooperation (e.g., coordination of ecologically sustainable horticulture); and iii) Investigating what the mechanisms are that make moralistic religious traditions more effective at expanding cooperation with unknown and distant co-religionists, if they are indeed more effective. Ultimately, answers to these questions will provide a more detailed and nuanced understanding of the role of supernatural punishment beliefs on human cooperation.

Acknowledgements

The authors gratefully acknowledge the hospitality of Yawari village and elders for this research. We also extend deep thanks to our onsite team coordinator Samuel Rodríguez and our lead translator Julie Rodríguez (no relation), as well as to Moises Castillo and Andrea Manotas for enthusiasm and professionalism in assistance during fieldwork.

Disclosure statement
The authors declare no conflicts of interest.

Authors' roles
CW and LC collected data, DM-S and BGP managed the project and contributed to analysis, and BGP contributed to the design. All authors contributed to writing this manuscript.

Funding
This project was made possible by a grant from the Templeton Religion Trust (#TRT-2022-31107). DM-S is also supported by the John Templeton Foundation (grant ref 61917).

Data availability statement
All data and analysis code are available on the project GitHub website: https://github.com/bgpurzycki/GGSL-Project/tree/main/Code/Objective%201/Kichwa.

A.I. statement
The authors declare that they did not use any A.I. in the design, execution, analysis, or writing of this paper.

References

Albrecht, S. L., & Heaton, T. B. (1984). Secularization, Higher Education, and Religiosity. *Review of Religious Research*, *26*(1), 43–58. https://doi.org/10.2307/3511041

Baimel, A., Apicella, C., Atkinson, Q., Bolyanatz, A., Cohen, E., Handley, C., Henrich, J., Lang, M., Lesorogol, C., Mathew, S., McNamara, R., Moya, C., Norenzayan, A., Placek, C., Soler, M., Weigel, J., Xygalatas, D., Ross, C., Vardy, T., … Purzycki, B. (2022). Material insecurity predicts greater commitment to moralistic deities and less commitment to local deities: A cross-cultural investigation. *Religion, Brain and Behavior*.

Baumard, N., & Boyer, P. (2013). Explaining moral religions. *Trends in Cognitive Sciences*, *17*(6), 272–280. https://doi.org/10.1016/j.tics.2013.04.003

Bendixen, T., Apicella, C., Atkinson, Q., Cohen, E., Henrich, J., McNamara, R., Norenzayan, A., Willard, A., Xygalatas, D., & Purzycki, B. G. (2024). Appealing to the minds of gods: Religious beliefs and appeals correspond to features of local social ecologies. *Religion, Brain & Behavior*, *14*(2), 183–205. https://doi.org/10.1080/2153599X.2023.2178487

Bendixen, T., Lightner, A. D., Apicella, C., Atkinson, Q., Bolyanatz, A., Cohen, E., Handley, C., Henrich, J., Klocová, E. K., Lesorogol, C., Mathew, S., McNamara, R. A., Moya, C., Norenzayan, A., Placek, C., Soler, M., Vardy, T., Weigel, J., Willard, A. K., … Purzycki, B. G. (2023). Gods are watching and so what? Moralistic supernatural punishment across 15 cultures. *Evolutionary Human Sciences*, *5*, e18. https://doi.org/10.1017/ehs.2023.15

Billingsley, J., Gomes, C. M., & McCullough, M. E. (2018). Implicit and explicit influences of religious cognition on Dictator Game transfers. *Royal Society Open Science*, *5*(8), 170238. https://doi.org/10.1098/rsos.170238

Blake, K. R., & Gangestad, S. (2020). On Attenuated Interactions, Measurement Error, and Statistical Power: Guidelines for Social and Personality Psychologists. *Personality and Social Psychology Bulletin*, *46*(12), 1702–1711. https://doi.org/10.1177/0146167220913363

Bozigar, M., Gray, C. L., & Bilsborrow, R. E. (2016). Oil Extraction and Indigenous Livelihoods in the Northern Ecuadorian Amazon. *World Development*, *78*, 125–135. https://doi.org/10.1016/j.worlddev.2015.10.035

Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, *80*, 1–28. https://doi.org/10.18637/jss.v080.i01

Chimbo-Gándara, L. F., Granda-Albuja, G., Mora, J. R., Llumiquinga, E., Ruiz-Uriguen, M., Machado, A., Cisneros-Heredia, D. F., Abreu-Naranjo, R., Giampieri, F., Tejera, E., & Álvarez-Suárez, J. M. (2024). Nutritional, functional, and safety characterization of the edible larva of the South American palm weevil (chontacuro) *Rhynchophorus palmarum* L. from Amazonian Ecuador. *Journal of Food Composition and Analysis*, *134*, 106507. https://doi.org/10.1016/j.jfca.2024.106507

Damian, C. (1995). *The Virgin of the Andes: Art and Ritual in Colonial Cuzco*. Grassfield Pr.

Davis, J., Sellers, S., Gray, C., & Bilsborrow, R. (2017). Indigenous migration dynamics in the

Ecuadorian Amazon: A longitudinal and hierarchical analysis. *The Journal of Development Studies*, *53*(11), 1849–1864. https://doi.org/10.1080/00220388.2016.1262028

Deffner, D., Fedorova, N., Andrews, J., & McElreath, R. (2024). Bridging theory and data: A computational workflow for cultural evolution. *Proceedings of the National Academy of Sciences*, *121*(48), e2322887121. https://doi.org/10.1073/pnas.2322887121

Durkheim, E. (2001). *The Elementary Forms of Religious Life*. Oxford University Press.

Erazo, J. S. (2013). *Governing Indigenous Territories: Enacting Sovereignty in the Ecuadorian Amazon*. Duke University Press.

Gomes, C. M., & McCullough, M. E. (2015). The effects of implicit religious primes on dictator game allocations: A preregistered replication experiment. *Journal of Experimental Psychology. General*, *144*(6), e94-104. https://doi.org/10.1037/xge0000027

Grzech, K., Schwarz, A., & Ennis, G. (2019). Divided we stand, unified we fall? The impact of standardisation on oral language varieties: a case study of Amazonian Kichwa. *Revista de Llengua i Dret*, 123–145. https://doi.org/10.2436/rld.i71.2019.3253

Henrich, J., Chudek, M., & Boyd, R. (2015). The Big Man Mechanism: How prestige fosters cooperation and creates prosocial leaders. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *370*(1683), 20150013. https://doi.org/10.1098/rstb.2015.0013

Heredia-R, M., Torres, B., Cayambe, J., Ramos, N., Luna, M., & Diaz-Ambrona, C. G. H. (2020). Sustainability Assessment of Smallholder Agroforestry Indigenous Farming in the Amazon: A Case Study of Ecuadorian Kichwas. *Agronomy*, *10*(12), 1973. https://doi.org/10.3390/agronomy10121973

Hernán, M. A., & Robins, J. (2020). *Causal Inference: What If*. Chapman & Hall/CRC.

Hruschka, D., Efferson, C., Jiang, T., Falletta-Cowden, A., Sigurdsson, S., McNamara, R., Sands, M., Munira, S., Slingerland, E., & Henrich, J. (2014). Impartial institutions, pathogen stress and the expanding social network. *Human Nature*, *25*(4), 567–579. https://doi.org/10.1007/s12110-014-9217-0

Ichinkhorloo, B., Major-Smith, D., & Purzycki, B. G. (2026). *Spirit-Master Beliefs and Cooperation Among the Buryat of Mongolia* (X2gzt_v1). SocArXiv. https://osf.io/preprints/socarxiv/x2gzt_v1/

Informe Social. (2003). *Informe Social 2003: Desarrollo Social y Pobreza en el Ecuador, 1990-2001*. Secretaría Técnica del Frente Social. https://web.archive.org/web/20111210160736/http://www.siise.gov.ec/Publicaciones/2inf9.pdf

Inquilla-Mamani, J., & Chambi-Apaza, E. (2019). Symbolic Communication and the Notion of "Pachamama" in the "Quechua" and "Aimara" Cultures. *International Journal of English Literature and Social Sciences*, *4*(2), 504–512.

Johnson, D. D. P. (2005). God's punishment and public goods: A test of the supernatural punishment hypothesis in 186 world cultures. *Human Nature*, *16*(4), 410–446.

https://doi.org/10.1007/s12110-005-1017-0
Johnson, D. D. P. (2015). *God Is Watching You: How the Fear of God Makes Us Human* (1 edition). Oxford University Press.
Johnson, D. D. P., & Krüger, O. (2004). The Good of Wrath: Supernatural Punishment and the Evolution of Cooperation. *Political Theology*, *5*(2), 159–176. https://doi.org/10.1558/poth.2004.5.2.159
Kohn, E. (2013). *How Forests Think: Toward an Anthropology Beyond the Human*. Univ of California Press.
Lang, M., Purzycki, B. G., Apicella, C. L., Atkinson, Q. D., Bolyanatz, A., Cohen, E., Handley, C., Kundtová Klocová, E., Lesorogol, C., Mathew, S., McNamara, R. A., Moya, C., Placek, C. D., Soler, M., Vardy, T., Weigel, J. L., Willard, A. K., Xygalatas, D., Norenzayan, A., & Henrich, J. (2019). Moralizing gods, impartiality and religious parochialism across 15 societies. *Proceedings of the Royal Society B: Biological Sciences*, *286*(1898), 20190202. https://doi.org/10.1098/rspb.2019.0202
Lang, M., Purzycki, B. G., Henrich, J., & Norenzayan, A. (2024). *The Evolution of Religion and Morality: Volume II*. Routledge.
Lightner, A. D., Bendixen, T., & Purzycki, B. G. (2023). Moralistic supernatural punishment is probably not associated with social complexity. *Evolution and Human Behavior, Special Issue: Dispatches from the Field Part II*, *44*(6), 555–565. https://doi.org/10.1016/j.evolhumbehav.2022.10.006
Lundberg, I., Johnson, R., & Stewart, B. M. (2021). What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory. *American Sociological Review*, *86*(3), 532–565. https://doi.org/10.1177/00031224211004187
Major-Smith, D. (2023). Exploring causality from observational data: An example assessing whether religiosity promotes cooperation. *Evolutionary Human Sciences*, *5*, e22. https://doi.org/10.1017/ehs.2023.17
Major-Smith, D., Morgan, J., Halstead, I., Tohidinik, H. R., Iles-Caven, Y., Golding, J., & Northstone, K. (2023). Demographic and socioeconomic predictors of religious/spiritual beliefs and behaviours in a prospective cohort study (ALSPAC) in Southwest England: Results from the parental generation. *Wellcome Open Research*, *7*, 159. https://doi.org/10.12688/wellcomeopenres.17897.3
Major-Smith, D., Morgan, J., Tohidinik, H. R., Goulding, N., Iles-Caven, Y., Golding, J., Northstone, K., & Halstead, I. (2023). Associations between psychological factors and religious/spiritual beliefs and behaviours in a prospective cohort study (ALSPAC) in Southwest England: A descriptive study. *Wellcome Open Research*, *8*, 174. https://doi.org/10.12688/wellcomeopenres.18955.2
Major-Smith, D., & Purzycki, B. G. (2026). Modeling Uncertainty around Free-list Cultural Salience Scores. *Field Methods*, *38*(1), 62–75. https://doi.org/10.1177/1525822X251379224
Malinowski, B. (1992). Malinowski and the Work of Myth. In I. Strenski (Ed.), *Malinowski and*

the Work of Myth. Princeton University Press. https://www.degruyter.com/document/doi/10.1515/9781400862801/html

Mateffy, A., Major-Smith, D., & Purzycki, B. G. (2026). *Spirit-Masters, Supernatural Punishment, and Religious Cooperation Among the Altai Uriankhai of Mongolia* (Enytz_v1). SocArXiv. https://osf.io/preprints/socarxiv/enytz_v1/

Minoia, P., Tapia, A., & Kaukonen Lindholm, R. E. (2024). Epistemic territories of kawsak sacha (living forest): Cosmopolitics and cosmoeducation. *Globalizations*, *21*(2), 331–348. https://doi.org/10.1080/14747731.2024.2308332

Naranjo, P. (1979). Hallucinogenic plant use and related indigenous belief systems in the Ecuadorian Amazon. *Journal of Ethnopharmacology*, *1*(2), 121–145. https://doi.org/10.1016/0378-8741(79)90003-5

Norenzayan, A. (2013a). *Big gods: How religion transformed cooperation and conflict*. Princeton University Press.

Norenzayan, A. (2013b). *Big gods: How religion transformed cooperation and conflict*. Princeton University Press.

Norenzayan, A., Shariff, A. F., Gervais, W. M., Willard, A. K., McNamara, R. A., Slingerland, E., & Henrich, J. (2016). The cultural evolution of prosocial religions. *Behavioral and Brain Sciences*, *39*, e1 (19 pages). https://doi.org/10.1017/S0140525X14001356

Norge, A., Major-Smith, D., & Purzycki, B. G. (2026). *The Cooperative Scope of Local Gods: Punishment, Omniscience, and Distant Cooperation among the Sihanaka of Madagascar* (Agnwe_v1). SocArXiv. https://osf.io/preprints/socarxiv/agnwe_v1/

Oikonomakis, L. (2024). Protecting the Forest Beings that Protect Us: The Cosmo-Political Challenge Kawsak Sacha Poses to Ecuador's Extraction-Based Development. *Bulletin of Latin American Research*, *43*(2), 145–158. https://doi.org/10.1111/blar.13535

Perreault, T. (2005). Why Chacras (Swidden Gardens) Persist: Agrobiodiversity, Food Security, and Cultural Identity in the Ecuadorian Amazon. *Human Organization*, *64*(4), 327–339.

Purzycki, B. G. (2011). Tyvan cher eezi and the socioecological constraints of supernatural agents' minds. *Religion, Brain & Behavior*, *1*(1), 31–45. https://doi.org/10.1080/2153599X.2010.550723

Purzycki, B. G. (2025a). *Ethnographic free-list data: Management and analysis with examples in R*. Sage Publications.

Purzycki, B. G. (2025b). *Morality and the Gods*. Cambridge University Press.

Purzycki, B. G., Apicella, C., Atkinson, Q. D., Cohen, E., McNamara, R. A., Willard, A. K., Xygalatas, D., Norenzayan, A., & Henrich, J. (2016). Moralistic gods, supernatural punishment and the expansion of human sociality. *Nature*, *530*(7590), 327–330. https://doi.org/10.1038/nature16980

Purzycki, B. G., & Bendixen, T. (2024). Material insecurity and religiosity: A causal analysis. *Evolutionary Human Sciences*, *6*, e4. https://doi.org/10.1017/ehs.2023.29

Purzycki, B. G., Bendixen, T., Lightner, A. D., & Sosis, R. (2022). Gods, games, and the socioecological landscape. *Current Research in Ecological and Social Psychology*, *3*,

100057. https://doi.org/10.1016/j.cresp.2022.100057
Purzycki, B. G., Henrich, J., & Norenzayan, A. (2024). *The Evolution of Religion and Morality: Volume I*. Routledge.
Purzycki, B. G., & Jamieson-Lane, A. (2017). AnthroTools: An R package for cross-cultural ethnographic data analysis. *Cross-Cultural Research*, *51*(1), 51–74.
Purzycki, B. G., & Major-Smith, D. (2026). *Toward a Quantitative and Causal Anthropology of Religion (QCAR): Part I*.
Purzycki, B. G., & McNamara, R. A. (2016). An ecological theory of gods' minds. In H. De Cruz & R. Nichols (Eds.), *Cognitive Science of Religion and Its Philosophical Implications* (pp. 143–167). Continuum.
Purzycki, B. G., Willard, A. K., Klocová, E. K., Apicella, C., Atkinson, Q., Bolyanatz, A., Cohen, E., Handley, C., Henrich, J., Lang, M., Lesorogol, C., Mathew, S., McNamara, R. A., Moya, C., Norenzayan, A., Placek, C., Soler, M., Vardy, T., Weigel, J., … Ross, C. T. (2022). The moralization bias of gods' minds: A cross-cultural test. *Religion, Brain & Behavior*, *12*(1–2), 38–60. https://doi.org/10.1080/2153599X.2021.2006291
R Core Team. (2024). *R: A Language and Environment for Statistical Computing* [Computer software]. R Foundation for Statistical Computing.
Rappaport, R. A. (1999). *Ritual and Religion in the Making of Humanity*. Cambridge University Press.
Rohrer, J. M. (2018). Thinking Clearly About Correlations and Causation: Graphical Causal Models for Observational Data. *Advances in Methods and Practices in Psychological Science*, *1*(1), 27–42. https://doi.org/10.1177/2515245917745629
Rudel, T. K., Bates, D., & Machinguiashi, R. (2002). Ecologically Noble Amerindians? Cattle Ranching and Cash Cropping among Shuar and Colonists in Ecuador. *Latin American Research Review*, *37*(1), 144–159.
Salibová, D. (2020). Ayahuasca Ethno-tourism and its Impact on the Indigenous Shuar Community (Ecuador) and Western Participants. *Computer Languages, Systems & Structures*, *4*(107). https://doi.org/10.21104/CL.2020.4.05
Scheepers, P., Te Grotenhuis, M., & Van Der Slik, F. (2002). Education, Religiosity and Moral Attitudes: Explaining Cross-National Effect Differences. *Sociology of Religion*, *63*(2), 157–176. https://doi.org/10.2307/3712563
Shariff, A. F., Willard, A. K., Andersen, T., & Norenzayan, A. (2016). Religious Priming: A Meta-Analysis With a Focus on Prosociality. *Personality and Social Psychology Review: An Official Journal of the Society for Personality and Social Psychology, Inc*, *20*(1), 27–48. https://doi.org/10.1177/1088868314568811
Singh, M., Kaptchuk, T. J., & Henrich, J. (2021). Small gods, rituals, and cooperation: The Mentawai water spirit Sikameinan. *Evolution and Human Behavior*, *42*(1), 61–72. https://doi.org/10.1016/j.evolhumbehav.2020.07.008
Smith, J. J., & Borgatti, S. P. (1997). Salience counts-and so does accuracy: Correcting and updating a measure for free-list-item salience. *Journal of Linguistic Anthropology*, *7*,

208–209.
Stan Development Team. (2026). *Stan Reference Manual* (Version v2.38.0) [Computer software]. https://mc-stan.org
Suzina, A. C. (2023). Towards a circular theory of communication: The case of the Wayusa ritual of the traditional Kichwa people of Sarayaku. *Journal of Alternative & Community Media*, *8*(Digital and Material Forms of Community and Alternative Media), 145–167. https://doi.org/10.1386/jacm_00124_1
Swanson, G. E. (1960). *The Birth of the Gods: The Origin of Primitive Beliefs*. University of Michigan Press.
Swanson, T. D. (2009). Singing to Estranged Lovers: Runa Relations to Plants in the Ecuadorian Amazon. *Journal for the Study of Religion, Nature and Culture*, *3*(1), 36–65. (Historical inquiry; Qualitative Research; Linguistic Analysis; Literary Analysis; Amazon; Ecuador; contemporary). https://doi.org/10.1558/jsrnc.v3i1.36
Tobes, I., Carrillo-Moreno, C., Guarderas-Flores, L., Jácome-Negrete, I., & Velázquez-Cárdenas, Y. (2022). Ethnoichthyology and Ethnotaxonomy of the Kichwa Indigenous People of Arawanu (Arajuno), in the Ecuadorian Amazon. *Frontiers in Ecology and Evolution*, *10*. https://doi.org/10.3389/fevo.2022.826781
Townsend, C., Aktipis, A., Balliet, D., & Cronk, L. (2020). Generosity among the Ik of Uganda. *Evolutionary Human Sciences*, *2*, e23. https://doi.org/10.1017/ehs.2020.22
Uzendoski, M. (2005). *The Napo Runa of Amazonian Ecuador*. University of Illinois Press.
Uzendoski, M., & Calapucha-Tapuy, E. F. (2012). *The Ecology of the Spoken Word: Amazonian Storytelling and the Shamanism among the Napo Runa*. University of Illinois Press.
Uzendoski, M., & Whitten, N. (2014). From "Acculturated Indians" to "Dynamic Amazonian Quichua-Speaking Peoples. *Tipití: Journal of the Society for the Anthropology of Lowland South America*, *12*(1), 1–13. https://doi.org/10.70845/2572-3626.1176
van Buuren, S. (2018). *Flexible Imputation of Missing Data* (Second). Chapman & Hall/CRC. https://stefvanbuuren.name/fimd/
Van Elk, M., Matzke, D., Gronau, Q., Guang, M., Vandekerckhove, J., & Wagenmakers, E.-J. (2015). Meta-analyses are no substitute for registered replications: A skeptical perspective on religious priming. *Personality and Social Psychology*, 1365. https://doi.org/10.3389/fpsyg.2015.01365
Vardy, T., Moya, C., Placek, C. D., Apicella, C. L., Bolyanatz, A., Cohen, E., Handley, C., Kundtová Klocová, E., Lesorogol, C., Mathew, S., McNamara, S. A., Purzycki, B. G., Soler, M., Weigel, J. L., Willard, A. K., Xygalatas, D., Norenzayan, A., Henrich, J., Lang, M., & Atkinson, Q. D. (2022). The religiosity gender gap in 14 diverse societies. *Religion, Brain & Behavior*, *12*(1–2), 18–37. https://doi.org/10.1080/2153599X.2021.2006292
Varela, R. (2025). Pueblo Kichwa de Rukullakta: Información. *Pueblo Kichwa de Rukullakta*. https://pueblorukullakta.blogspot.com/p/informacion_30.html
Vasco, C., Bilsborrow, R., Torres, B., & Griess, V. (2018). Agricultural land use among mestizo

colonist and indigenous populations: Contrasting patterns in the Amazon. *PLoS ONE*, *13*(7), e0199518. https://doi.org/10.1371/journal.pone.0199518

Vera-Velez, R. R., Cota-Sánchez, J. H., & Grijalva Olmedo, J. E. (2019). Biodiversity, dynamics, and impact of chakras on the Ecuadorian Amazon. *Journal of Plant Ecology*, *12*(1), 34–44. https://doi.org/10.1093/jpe/rtx060

Volk, S., Thöni, C., & Ruigrok, W. (2011). Personality, personal values and cooperation preferences in public goods games: A longitudinal study. *Personality and Individual Differences*, *50*(6), 810–815. https://doi.org/10.1016/j.paid.2011.01.001

White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, *30*(4), 377–399. https://doi.org/10.1002/sim.4067

Whitten, N. E., & Whitten, D. S. (2008). *Puyo Runa: Imagery and Power in Modern Amazonia*. University of Illinois Press.

Varela, R. (2025). Pueblo Kichwa de Rukullakta: Información. *Pueblo Kichwa de Rukullakta*. https://pueblorukullakta.blogspot.com/p/informacion_30.html

Vasco, C., Bilsborrow, R., Torres, B., & Griess, V. (2018). Agricultural land use among mestizo colonist and indigenous populations: Contrasting patterns in the Amazon. *PLoS ONE*, *13*(7), e0199518. https://doi.org/10.1371/journal.pone.0199518

Vera-Velez, R. R., Cota-Sánchez, J. H., & Grijalva Olmedo, J. E. (2019). Biodiversity, dynamics, and impact of chakras on the Ecuadorian Amazon. *Journal of Plant Ecology*, *12*(1), 34–44. https://doi.org/10.1093/jpe/rtx060

Whitten, N. E., & Whitten, D. S. (2008). *Puyo Runa: Imagery and Power in Modern Amazonia*. University of Illinois Press.